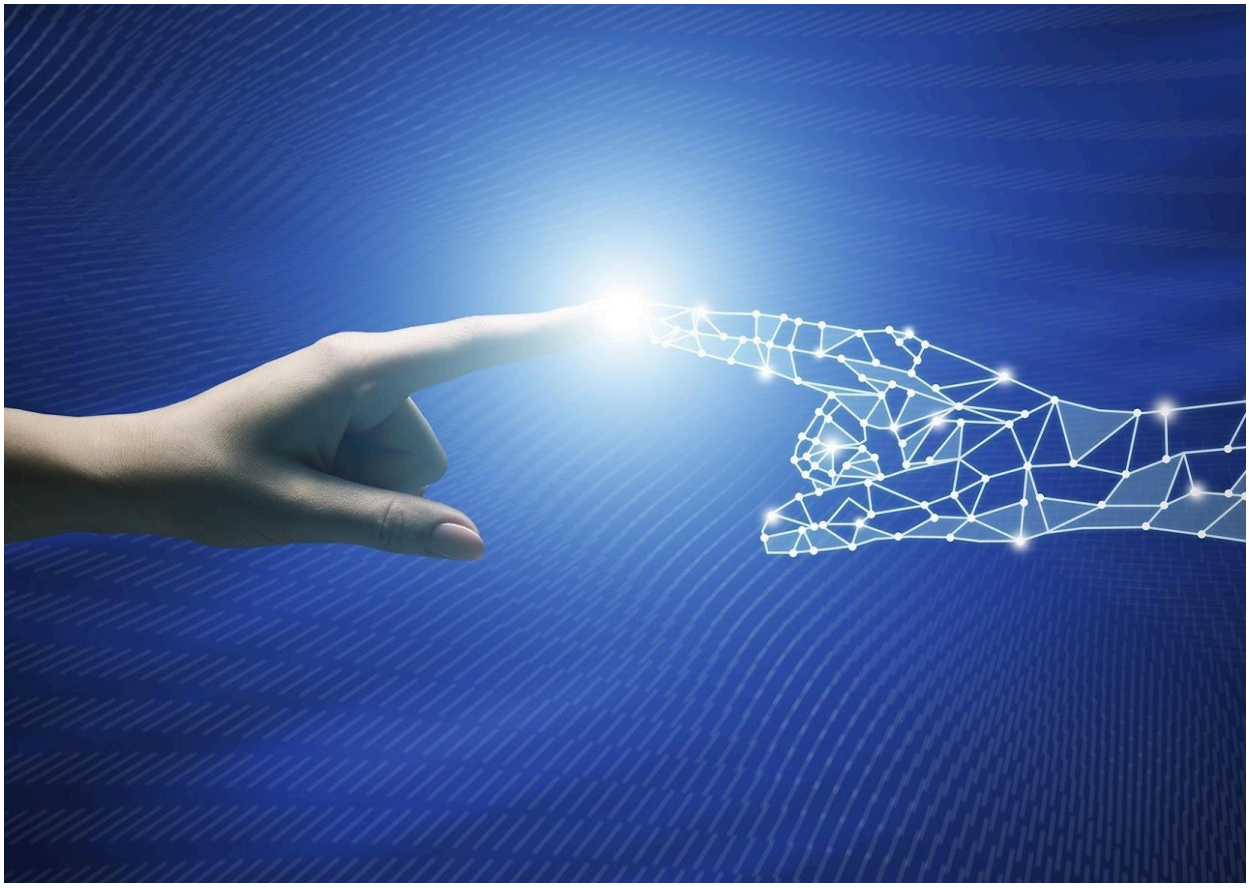
Machine Learning Document

Document containing fundamentals learnt during machine learning



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Models:

XgBoost -

Components - gradient boosting + decision tree ie. example of ensemble model.

Advantages - not sensitive to multicollinearity, learns non-linear relationships between features well so that you do not have to explicitly feature engineer complex relationships. Works well with parallel processing. Not affected by outliers within the data set. Produces MI scores that form a great basis for feature engineering.

Disadvantages - tends to be highly accurate and may lead you to spurious accuracy. Long training time. Hard to interpret for non-technical members.

Overview: A model that combines gradient boosting and decision trees. Trees are developed based on features and splits occur depending on gain, the sum of the first and second derivative of the loss function. Gradient boosting computes a weak predictor value then gets the error from the prediction and trains subsequent models on residuals then finally makes a prediction by adding up all the errors plus the initial weak prediction.

Illustration: makes a weak prediction with random features -> uses the error of the predictions to train a new small model -> makes a small decision tree model to predict on errors -> continues until error meets criteria.

Key parameters:

learning rate - controls that rate at which the model changes losses during stochastic gradient descent. A larger value i.e. 1 corresponds with the model underfitting. A smaller value i.e. 0.01 corresponds with the model overfitting in rare cases.

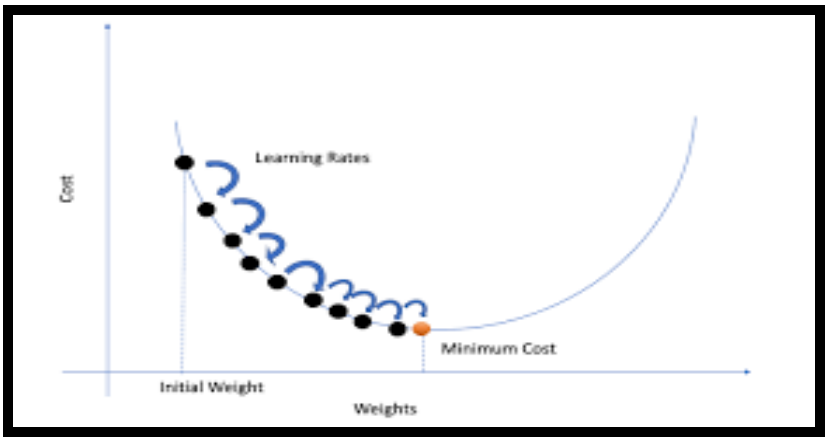

Image by ResearchGate: https://share.google/7sSVSuaaY5INcD9e5

If you have a larger learning rate, your model is more likely to ‘over-step’ the minimum loss.

verbose - Number that tells python how many messages to print during training

n_estimators - number of iterations the model performs. Higher values correspond to overfitting, lower values correspond to underfitting.

early_stopping_rounds - a positive integer value that tells the model when it’s overfitting i.e. when the validation score stops improving. Goes hand in hand with n_estimators and the eval_set parameter.

eval_set - a comparison data set to score the performance of the model on data unseen. Usually your test training data eg. eval_set = [(Xtrain,ytrain)]

n_jobs - value either 1 or -1. The former doesn’t use all your pc resources, the latter uses all your computer resources to speed up the process

max_depth - the maximum number of nodes developed by a tree, the more nodes, the deeper the thinking and more complex relationships found. Note that larger values correspond to overfitting and are computationally exhaustive. Natural numbers

min_child_weight - numeric value that controls how many complex relationships the model learns in each node. Smaller values correspond to more complexity, larger values correspond to less. Useful in cases of small data sets to prevent overfitting. Natural numbers

colsample_bytree - fraction of features presented in each tree. Introduces randomness and correlation between trees to prevent overfitting.(‘Xgboosting.com’). Rational numbers.

reg_alpha - encourages sparse solutions and prevents overfitting. Larger value of alpha, reduce overfitting. Any positive rational value. Uses a penalty that reduces some weights to zero.

reg_lambda - similar to reg_alpha but smoothens out weights.

Scale_pos_weight - motivates your model to search for more of the less frequent class. Formula = sum(0)/sum(1)

Key metrics for classification:

Look at the formula of metrics, this tells you what we want to optimize.

Precision - tp/(tp+fp) here fp appears in the formula, cases where fp's are a huge risk, we focus on precision. Adjust by changing the threshold for probabilities. Larger threshold -> fewer false positives -> higher precision

Recall - tp/(tp+fn) here fn appears in the formula, cases where fn's are a huge risk, we want to focus more on recall

Random Forest-

A decision tree algorithm. First the model creates a sample of N. Selects m features, splits nodes depending on metric then finally makes a prediction on either the mode of each tree for classification or the mean for each tree based on regression. Useful features are ones that ease the splitting at each tree.

Assumption: Random forest requires enough data

Hyperparameters:

n_estimators - number of decision trees, the higher the value, the better the predictions but longer training time. Positive integer value

max_features - maximum number of features at each node. Positive integer value

max_depth - The amount of non-linear interactions you want the tree to discover. Longer training times and risk of overfitting. Positive integer. Default is none and then depends on min_samples_split.

min_samples_split - Minimum number of samples required for the tree to make a split. Prevents overfitting on small sample sizes. Positive integer value.

min_samples_leaf - Controls the number of samples at each leaf. Higher values prevent overfitting. Positive integer value.

MLPClassifier-

Neural network that works best on non-linear relationships between features.

Neural networks operate like our brains. Nodes are seen as where transformations occur. Makes use of forward and back propagation to adjust weights behind predictions. The more layers, the more complex relationships found within data. The drawback is the model tends to overfit.

Advantages - Tends to not be overly accurate. Great practical use. Abundance of online resources and models to use for transfer learning to solve similar problems.

Disadvantages - Does not produce prediction probabilities. Requires a lot of processing power. The model learns from tensors and not actual tabular data which decreases interpretability behind model findings.

Hyperparameters:

hidden_layer_size - specifies number of hidden layers and neurons. More neurons and networks means more complex relationships found within the model.

max_iter - number of epochs

activation function - functions that transform the data from neuron to neuron, common functions include: tanh, relu, sigmoid, softmax

learning rate - how much the model adjust loss, larger values lead to underfitting, lower values lead to overfitting

solver - also known as optimizer, how the model tries to solve loss issue. Common solvers include, Adam, SGDand lbfgs

Neural networks from scratch using torch:

hyperparameters:

input_features - the number of features accepted at each node in the learning process. Nodes learn surface level information.

output_features - the required output for the model, for classification and

regression usually 1 and for image recognition depends on recognition task and metrics of images.

Credit to [Daniel Bourke](#) for his vast array of free online neural network information.

Light gradient boosting -

Ensemble method similar to xgboost. The algorithm uses histogram based learning i.e. turning continuous features into bins -> splits nodes at maximum gain -> samples features from samples with small gradient -> builds it on features.

Advantages - similar to xgboost but just not as flexible and robust.

Disadvantage - also leads to spurious accuracy

Hyperparameters:

learning_rate

num_iterations - similar to n_estimators

objective - binary or multiclass or regression

max_depth

num_leaves -

AdaBoost

Advantage of AdaBoost is it's not prone to overfitting but a disadvantage is it may underfit the data. Algorithm: random training data -> trains model -> assigns weights depending on accuracy -> repeats

Hyperparameters:

base_estimator - the learning algorithm used to train weak models

n_estimators - number of iterations

learning_rate